

02_VEHICLE_FIRMWARE_PARITY

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EVIDENCE PROVENANCE: [FIRMWARE AUDIT]
Side-by-side verification of Vehicle A (sketch_aug26a.ino) and Vehicle B firmware source code.

1. Common Protocol Contracts

- **Unified Sequence:** Both vehicles maintain monotonic globalSequence across Wi-Fi and LoRa transports.
- **Persistent Boot ID:** ESP32 NVS Flash (Preferences.h) increments bootId on reboot to prevent out-of-order false alarms.
- **Safe Beacon Fallback:** Both vehicles autonomously trigger 1.0 Hz LoRa beaconing upon Wi-Fi drop.

2. Hardware Parity Matrix

Subsystem	Vehicle A (TRUCK_01)	Vehicle B (TRUCK_02)	Status
V2V Transport	433 MHz LoRa + Wi-Fi	433 MHz LoRa + Wi-Fi	PARITY
V2V Wire Format	11 comma-separated fields	11 comma-separated fields	PARITY
TDM Time Slot	500 ms in 2000 ms frame	1000 ms in 2000 ms frame	PARITY
Safe Beacon	1.0 Hz on Wi-Fi drop	1.0 Hz on Wi-Fi drop	PARITY
Physical Slots (PPR)	42 slots	43 slots	HARDWARE
Calibration K_cal	34.58 pulses/rev	34.58 pulses/rev	PARITY
Motor Driver	L298N Bipolar	TB6612FNG MOSFET + STBY	HARDWARE
Startup Safety	Zero PWM at boot	STBY LOW, Zero PWM	PARITY